SRF-UNet: A Lightweight Retinal Vessel Segmentation Network Based on Serial Residual Fusion and Bipolar Interaction

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***Abstract*—Accurate retinal vessel segmentation supports ophthalmic screening and vascular morphology analysis, but thin branches, low contrast, lesions, and uneven illumination often produce discontinuities and background false positives. This paper presents SRF-UNet, a compact five-stage encoder-decoder composed of three complementary modules. Serial residual fusion depthwise convolution (SRFDC) progressively models multiple receptive fields; serial residual fusion attention (SRFA) combines directional Coordinate Attention with the SRFDC path; and bipolar cross-interaction fusion (BCIF) regulates skip information according to encoder-decoder agreement and discrepancy. Experiments on DRIVE, STARE, and CHASE_DB1 show that the base model uses 0.34 M parameters and 0.26 G FLOPs while obtaining considerable Dice scores. The results demonstrate a favorable balance among segmentation quality, centerline continuity, and computational efficiency. Code is available at https://github.com/illuminatewang/SRF-UNet.**

Keywords*—*lightweight convolutional neural network, retinal vessel segmentation, multi-scale feature fusion

I. INTRODUCTION

Retinal vessels provide clinically useful evidence for diabetic retinopathy, hypertension, and cardiovascular disease. Automated segmentation supports vessel-width and centerline analysis. However, capillaries occupy only a small image region and frequently exhibit low contrast, tortuous geometry, lesion interference, and uneven illumination, making it difficult to preserve continuous thin branches while suppressing background responses.

U-Net[1] established the encoder-decoder paradigm, and UNet++[2] reduces the cross-level semantic gap through nested skips. SA-UNet[3] and FR-UNet[4] improve spatial attention and high-resolution interaction, but dense or full-resolution exchange increases memory and computation. G-Net-Light[5], LMBiS-Net[6], and LFA-Net[7] pursue lightweight inference, yet conventional skip connections can still transfer lesion texture and illumination artifacts without checking whether encoder and decoder features agree. TransUNet[8] and Mamba[9] extend global context modeling, while UNeXt[15], Swin-Unet[16], U-Mamba[17], and Swin-UMamba[18] adapt lightweight, Transformer, or state-space designs to medical segmentation; however, token interaction or directional scanning can complicate small-model deployment. Coordinate Attention[10] provides a lighter directional alternative, and clDice[11] reveals centerline breaks that may be hidden by region-overlap metrics.

SRF-UNet addresses three linked problems: progressive receptive-field construction, direction-aware enhancement, and selective cross-level transfer. The main contributions are:

- SRFDC uses three serial 3×3 depthwise convolutions, stage-wise residuals, and pixel-adaptive fusion weights to capture capillary texture and broader context at low cost.
- SRFA combines Coordinate Attention[10] with an SRFDC inverted-residual path in both encoder and decoder, preserving height- and width-sensitive position during representation and reconstruction.
- BCIF jointly models encoder-decoder agreement and absolute discrepancy, producing a bipolar gate that suppresses, preserves, or enhances skip information according to cross-level consistency.

II. METHOD

A. Network Overview

Let the input be $I \in \mathbb{R}^{B\times C_{\text{img}}\times H_0\times W_0}$. SRF-UNet uses five encoding levels and a symmetric decoder. The channel vectors are $[8,16,32,48,80]$, $[16,32,64,96,160]$, and $[32,64,128,192,320]$ for SRF-UNet-S, SRF-UNet, and SRF-UNet-L, respectively. Figure 1 presents the complete architecture and the internal structures of SRFA, SRFDC, and BCIF. The encoder reduces spatial resolution, the decoder restores it by bilinear upsampling, and same-scale features pass through BCIF rather than direct concatenation. The figure clarifies that SRFDC supplies context inside SRFA, SRFA is reused in both paths, and BCIF governs their information exchange.

B. Serial Residual Fusion Depthwise Convolution

For a channel-expanded input $Z_0 \in \mathbb{R}^{B\times C_e\times H\times W}$ with $C_e = 2C_{\text{in}}$, SRFDC constructs progressive context through three serial depthwise stages:

$$Z_i = Z_{i-1} + \mathrm{ReLU6}\big(\mathrm{BN}_i(\mathrm{DWConv}^{i}_{3\times3}(Z_{i-1}))\big). \quad (1)$$

The stages Z_1, Z_2, and Z_3 approximate 3×3, 5×5, and 7×7 effective receptive fields. A pointwise layer predicts spatial scores:

$$[Q_1, \dots, Q_m] = \mathrm{Conv}^{mC_e\to m}_{1\times1}([Z_1; \dots ; Z_m]). \quad (2)$$

The normalized contribution of stage i is

$$\alpha_i = m\frac{\exp(Q_i)}{\sum_{j=1}^{m}\exp(Q_j)}. \quad (3)$$

The fused output is

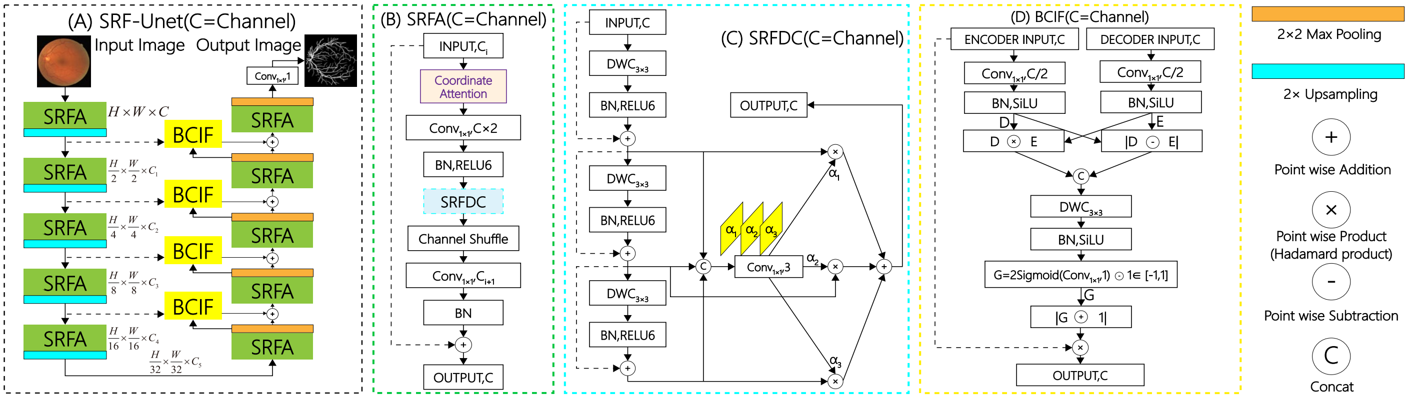

Figure 1. The proposed SRF-UNet architecture. (a) Five-stage SRF-UNet encoder-decoder, (b) serial residual fusion attention (SRFA), (c) serial residual fusion depthwise convolution (SRFDC), and (d) bipolar cross-interaction fusion (BCIF). C1–C5 denote the channel widths of the five encoder stages; [C1,C2,C3,C4,C5] is set to [8,16,32,48,80], [16,32,64,96,160], and [32,64,128,192,320] for SRF-UNet-S, SRF-UNet, and SRF-UNet-L, respectively. CA denotes Coordinate Attention[10]; DWC denotes depthwise convolution; BN denotes batch normalization.

$$\mathrm{SRFDC}(Z_0) = \textstyle\sum_{i=1}^{m} \alpha_i \odot Z_i. \quad (4)$$

The zero-initialized score layer initially approximates equal weighting and then learns pixel-wise scale selection. Local residuals preserve the response of narrow vessels even when later stages aggregate wider context.

C. Serial Residual Fusion Attention

SRFA is shared by the encoder and decoder. Coordinate Attention[10] first preserves direction-aware position:

$$X_{\mathrm{CA}} = \mathrm{CA}(X) = X \odot A_h \odot A_w. \quad (5)$$

With expansion $\mathcal{E}$, shuffled output projection $\mathcal{P}$, and residual mapping R, the block output is

$$\mathrm{SRFA}(X) = R(X) + \mathcal{P}(\mathrm{SRFDC}(\mathcal{E}(X_{\mathrm{CA}}))). \quad (6)$$

The residual mapping is

$$R(X) = \begin{cases} X, & C_{\mathrm{in}} = C_{\mathrm{out}}, \\ \mathrm{Conv}_{1\times 1}(X), & C_{\mathrm{in}} \neq C_{\mathrm{out}}. \end{cases} \quad (7)$$

A lightweight expansion ratio and channel shuffling reduce feature-transformation cost. This design retains an identity route for stable optimization while the attention-guided SRFDC branch emphasizes elongated, directionally coherent vessel structures.

D. Bipolar Cross-Interaction Fusion

BCIF projects same-scale decoder and encoder features $g, x \in \mathbb{R}^{B\times C\times H_i\times W_i}$ into a shared $C_f = C/2$ subspace:

$$D = \mathrm{SiLU}(\mathrm{BN}(\mathrm{Conv}_{1\times 1,d}(g))). \quad (8)$$

$$E = \mathrm{SiLU}(\mathrm{BN}(\mathrm{Conv}_{1\times 1,e}(x))). \quad (9)$$

Agreement and discrepancy are concatenated as

$$F_{\mathrm{int}} = [D \odot E; |D - E|]. \quad (10)$$

The interaction is converted into a pixel-wise bipolar gate:

$$A = 2\sigma(\mathrm{Conv}_{1\times 1}(\mathrm{SiLU}(\mathrm{BN}(\mathrm{DWConv}_{3\times 3}(F_{\mathrm{int}}))))) - 1. \quad (11)$$

Finally,

$$\mathrm{BCIF}(g, x) = g + x \odot (1 + A). \quad (12)$$

Because $A \in (-1,1)$, the gate can suppress inconsistent skip responses, preserve neutral evidence, or enhance mutually supported vessel features. Zero initialization makes the initial fusion close to $g + x$, preventing unstable early gating.

E. Complete Network and Objective

At encoder level i, SRFA produces

$$E_i = \mathrm{SRFA}_i^{e}(X_{i-1}). \quad (13)$$

The skip feature is obtained by

$$T_i = \mathrm{MaxPool}_{2\times 2}(E_i). \quad (14)$$

The decoder applies SRFA, bilinear upsampling, and BCIF:

$$V_i = \mathrm{BCIF}_i(\mathrm{Up}_{\times 2}(\mathrm{SRFA}_i^{d}(V_{i+1})), T_i). \quad (15)$$

A 1×1 head and sigmoid generate the vessel probability map. Training uses the region-aware structure loss adopted by PraNet[19]. A local pooling difference emphasizes boundaries and thin structures:

$$w_i = 1 + 5|\mathrm{AvgPool}_{31\times 31}(y)_i - y_i|. \quad (16)$$

The weighted binary cross-entropy term is

$$\mathcal{L}_{\mathrm{wBCE}} = -\frac{\sum_i w_i[y_i \ln p_i + (1-y_i)\ln(1-p_i)]}{\sum_i w_i}. \quad (17)$$

The weighted IoU term is

$$\mathcal{L}_{\mathrm{wIoU}} = 1 - \frac{\sum_i w_i p_i y_i + \varepsilon}{\sum_i w_i (p_i + y_i - p_i y_i) + \varepsilon}. \quad (18)$$

The final objective is

$$\mathcal{L}_{\mathrm{seg}} = \mathcal{L}_{\mathrm{wBCE}} + \mathcal{L}_{\mathrm{wIoU}}. \quad (19)$$

Here p_i and y_i denote prediction and label, and $\varepsilon = 1$. clDice is used only for evaluation and does not participate in backpropagation.

III. EXPERIMENTS AND RESULTS

A. Datasets, Training, and Metrics

Experiments use DRIVE[12], STARE[13], and CHASE_DB-1[14] with fixed 20/20, 15/5, and 20/8 training-test splits. RGB images are normalized with ImageNet statistics and sampled online as 224×224 patches with stride 112. Random $\pm 90°$ rotation and independent horizontal/vertical flips are applied. Testing retains the original resolution; overlapping probability patches are blended with two-dimensional Gaussian weights and thresholded at 0.5 to reduce seam artifacts.

All models are trained for 200 epochs with batch size 8. AdamW uses an initial learning rate of 5×10^{-4} and weight decay 10^{-4}, while cosine annealing reduces the rate to 10^{-6}. Gradients are clipped to $[-0.5, 0.5]$. Convolution weights are initialized from a normal distribution with standard deviation 0.02, and BatchNorm scale and bias are initialized to one and zero. Implementation uses PyTorch 2.11+cu128 on an NVIDIA GeForce RTX 5070 Ti. Results are reported as mean ± standard deviation over seeds 2026, 2027, and 2028. Dice measures regional overlap, clDice [11] measures centerline continuity, Acc measures pixel classification, and inference Throughput (images/s) measures model execution speed at batch size 16. Params

Table 1. Comparison with state-of-the-art methods on DRIVE, STARE, and CHASE_DB1. Results are reported as mean ± standard deviation over three runs, where the value after ± denotes the standard deviation in percentage points. The best mean scores are highlighted in bold, and the second-best mean scores are underlined.

Network	Year	Params (M)	FLOPs (G)	Throughput (images/s)	Performance Measures (%)								
					DRIVE			STARE			CHASE_DB1		
					Dice	clDice	Acc	Dice	clDice	Acc	Dice	clDice	Acc
U-Net[1]	2015	31.03	41.90	352	80.53±0.11	82.60±0.86	96.41±0.06	81.50±0.31	84.26±0.20	97.11±0.05	81.69±0.41	86.54±0.24	96.55±0.08
UNet++[2]	2018	8.57	26.30	338	80.64±0.13	82.82±0.73	96.36±0.05	81.73±0.35	84.82±0.28	97.32±0.07	81.97±0.43	86.42±0.31	97.13±0.07
SA-UNet[3]	2020	0.53	2.41	2524	80.38±0.09	83.03±0.31	96.28±0.04	80.41±0.61	85.04±0.29	96.92±0.13	81.44±0.34	86.38±0.36	97.30±0.05
FR-UNet[4]	2022	5.72	45.15	158	79.73±0.19	83.69±0.42	96.01±0.08	81.93±0.55	86.37±0.46	97.08±0.09	81.91±0.27	86.31±0.29	97.36±0.06
G-Net-Light[5]	2022	0.36	8.51	579	80.74±0.09	83.72±0.31	96.39±0.07	80.52±0.29	84.45±0.42	96.98±0.04	81.26±0.2	84.79±0.46	97.34±0.08
LMBiS-Net[6]	2024	0.17	1.22	2111	80.55±0.13	83.76±0.27	96.31±0.05	82.69±0.61	86.42±0.63	97.39±0.06	82.41±0.21	86.44±0.33	97.39±0.06
LFA-Net[7]	2025	0.11	0.64	**4453**	79.79±0.11	83.85±0.23	96.11±0.07	78.64±0.31	81.89±0.43	96.64±0.05	80.54±0.24	84.89±0.36	97.19±0.05
SRF-UNet-S	-	**0.10**	**0.10**	1633	80.16±0.08	83.13±0.14	96.32±0.06	81.83±0.28	84.71±0.31	97.20±0.06	81.83±0.26	85.59±0.32	97.40±0.04
SRF-UNet	-	0.34	0.26	1256	80.79±0.06	83.84±0.24	96.43±0.06	82.71±0.26	**86.79±0.26**	97.42±0.04	82.52±0.15	86.87±0.27	97.49±0.06
SRF-UNet-L	-	1.21	0.79	510	**80.94±0.07**	**84.10±0.13**	**96.46±0.02**	**82.80±0.17**	86.68±0.24	**97.47±0.03**	**82.90±0.17**	**86.89±0.10**	**97.61±0.04**

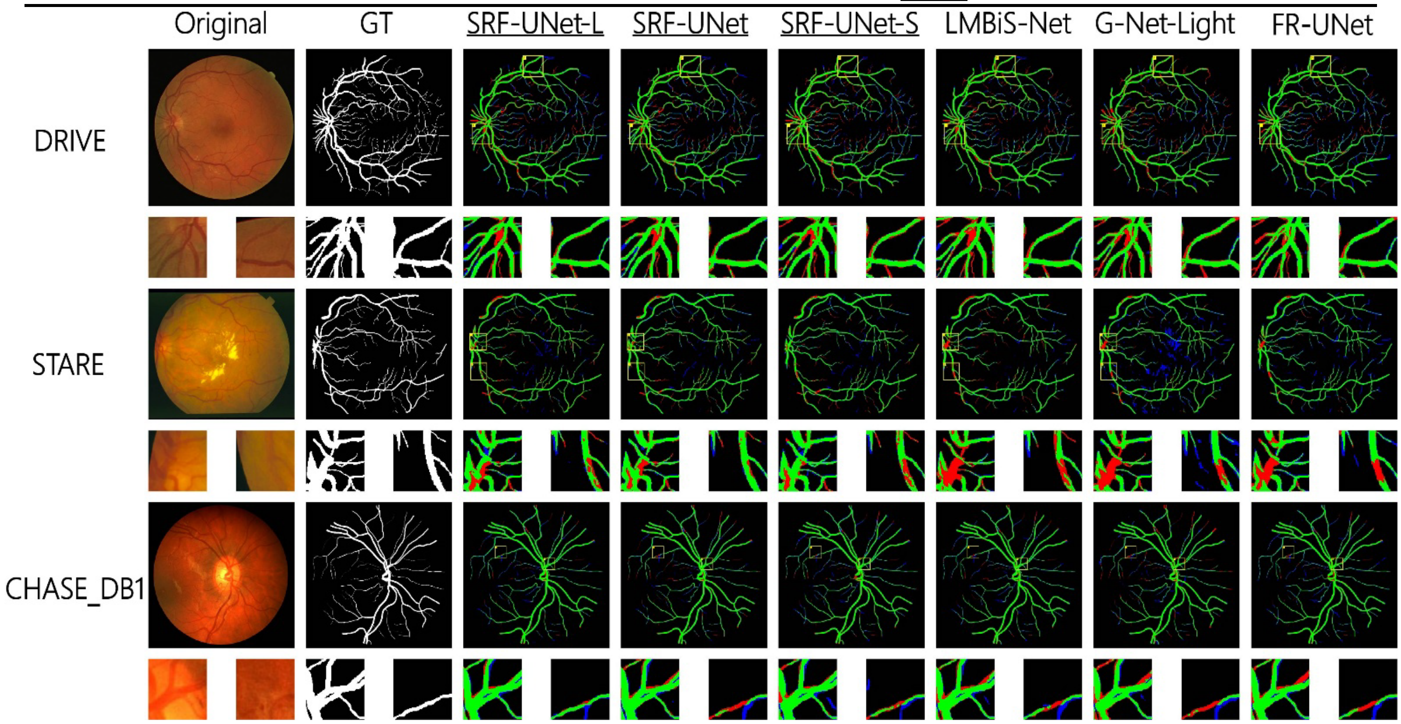

Figure 2. Qualitative comparison of retinal vessel segmentation on DRIVE, STARE, and CHASE_DB1. Each group shows the original image, ground truth (GT), and predictions from SRF-UNet-L, SRF-UNet, SRF-UNet-S, LMBiS-Net [6], G-Net-Light [5], and FR-UNet [4]. Green, blue, and red denote true-positive, false-positive, and false-negative vessel pixels, respectively; black denotes correctly classified background. Yellow boxes mark the enlarged regions.

and FLOPs describe model size and forward computation.

B. Quantitative Results

Table 1 compares SRF-UNet with representative accurate and lightweight networks on all three datasets. SRF-UNet-S uses only 0.10 M parameters and 0.10 G FLOPs, establishing a compact operating point. With 0.34 M parameters and 0.26 G FLOPs, the base SRF-UNet achieves 80.79%, 82.71%, and 82.52% Dice on DRIVE, STARE, and CHASE_DB1, respectively, while reducing FLOPs by 96.94% relative to G-Net-Light [5] and by 78.69% relative to LMBiS-Net [6]. Its throughput reaches 1,256 images/s, exceeding U-Net, UNet++, FR-UNet, and G-Net-Light, although SA-UNet, LMBiS-Net, and LFA-Net run faster. Notably, the fastest model, LFA-Net, trails the base SRF-UNet in Dice by 1.00, 4.07, and 1.98 percentage points on the three datasets. On STARE, SRF-UNet obtains 86.79% clDice and 97.42% Acc, while it reaches 86.87% clDice and 97.49% Acc on CHASE_DB1. SRF-UNet-L achieves the best Dice on all three datasets with only 0.79 G FLOPs. These results demonstrate a favorable balance among segmentation accuracy, model complexity, and inference throughput rather than peak throughput alone.

C. Qualitative Results

Figure 2 complements Table 1 by comparing vessel maps and enlarged local regions. Green, blue, and red pixels denote true positives, false positives, and false negatives. On DRIVE, SRF-UNet and SRF-UNet-L retain continuous capillaries near the optic disc and reduce missed endpoints. On STARE, they suppress lesion-related responses while preserving low-contrast connections. On CHASE_DB1, they recover dense peripheral vessels with fewer false links. The remaining errors concentrate around faint terminals and ambiguous lesion boundaries. The evidence is consistent with SRFDC context aggregation, SRFA directional enhancement, and BCIF consistency modeling.

D. Ablation Results

Table 2 evaluates the modules on the same five-stage depthwise U-Net backbone. SRFDC, SRFA, and BCIF individually increase Dice by 0.43, 0.28, and 0.32 points and clDice by 0.54, 0.43, and 0.55 points. BCIF provides the largest single-module topological gain. Every two-module setting exceeds the baseline, and SRFDC+SRFA is the strongest partial model. The complete network improves Dice, clDice, and Acc by 1.04, 1.30, and 0.56 points for only 0.07 M additional parameters and 0.08 G FLOPs, showing that the three modules address complementary error sources.

Table 2. Ablation study of the proposed modules on the DRIVE dataset. Results are reported as mean ± standard deviation over three runs, where the value after ± denotes the standard deviation in percentage points. The best values are shown in bold.

Variant	SRFDC	SRFA	BCIF	Dice	clDice	Acc	Params/M	FLOPs/G
Baseline (DWC-UNet)	×	×	×	79.75±0.30	82.54±0.34	95.87±0.06	0.27	0.18
Baseline+SRFDC	✓	×	×	80.18±0.18	83.08±0.15	96.09±0.08	0.31	0.24
Baseline+SRFA	×	✓	×	80.03±0.25	82.97±0.39	96.01±0.04	0.28	0.18
Baseline+BCIF	×	×	✓	80.07±0.18	83.09±0.20	96.06±0.06	0.29	0.19
Baseline+SRFA+BCIF	×	✓	✓	80.58±0.17	83.68±0.30	96.34±0.10	0.30	0.20
Baseline+SRFDC+BCIF	✓	×	✓	80.55±0.24	83.58±0.18	96.35±0.07	0.29	0.20
Baseline+SRFDC+SRFA	✓	✓	×	80.63±0.10	83.78±0.23	96.38±0.04	0.32	0.24
SRF-UNet	✓	✓	✓	**80.79±0.06**	**83.84±0.24**	**96.43±0.06**	0.34	0.26

E. Discussion and Limitations

The results demonstrate that SRF-UNet achieves efficient retinal vessel segmentation through coordinated feature modeling rather than channel-width reduction alone. SRFDC enlarges the effective receptive field, SRFA introduces directional location cues, and BCIF selectively attenuates or emphasizes same-scale encoder information. Their complementary roles support consistent Dice and clDice improvements, particularly on the challenging STARE and CHASE_DB1 datasets. Although SRF-UNet does not achieve the highest throughput, the base model processes 1,256 images/s while substantially outperforming the fastest model, LFA-Net, in Dice across all three datasets. This result highlights a favorable balance among segmentation accuracy, computational complexity, and inference speed. Moreover, SRF-UNet-L achieves the best Dice scores with only 0.79 G FLOPs, demonstrating that the architecture can be scaled effectively without changing its core design. The main limitations lie in the relatively small size of the dataset and the insufficient research on cross-device domain shifts. These issues will be addressed in future work.

IV. CONCLUSION

SRF-UNet combines serial residual features, coordinate-aware attention, and bipolar cross-level interaction in a regular five-stage encoder-decoder. The architecture preserves overlap and vessel topology with 0.10–0.79 G FLOPs across three model scales. Quantitative, qualitative, and ablation evidence shows that the three modules jointly improve thin-vessel continuity, reduce lesion-induced false positives, and maintain a favorable accuracy-efficiency balance. Future work will study cross-device domain shifts, calibration under image-quality degradation, and measured latency on embedded hardware.

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